

North Star (Multi-Source Candidate Data Transformer) - Technical Design

1. PROBLEM FRAMING:

Candidate data is collected from many disparate sources (ATS exports, recruiter files, resumes, notes, and profiles), in many different formats with duplication and conflicting data. The objective is to **consolidate** this difficult multi-source data into a single candidate profile with a focus on **provenance, explainability, and deterministic behavior**. The system must adopt a strategy of "**wrong-but-confident is worse than honestly-empty**" to resolve conflicts while ensuring conservative resolutions to ensure no false merges occur. The system ought to preserve **traceability** of outputs to their original data.

2. PIPELINE & ARCHITECTURE

- Evidence-first core:** every fact must be an Evidence atom (field_path, value, raw_value, source_id, source_type, method, normalized, order_index). Evidence merging is a pure function defined over a sorted evidence set, complete provenance, structural determinism, and I/O-free unit testing.
- Strict boundary:** Stages 1-6 extract and merge sources (without consideration for the output schema), while Stages 7-8 are pure transformations (canonical_record, config) to the output (without consideration for the sources). Default and custom outputs rely on the same validation pipeline.
- Deterministic extraction:** we extract (with regex, alias dictionaries and grammars) structured (CSV or JSON), unstructured (PDF or DOCX), and URL-keyed (GitHub or LinkedIn) sources, and provided notes. No live LLMs are used, thus every run is fully reproducible.
- Fault isolation:** adapters are placed within fault boundaries, sources that become corrupted should be logged as skipped / partial without stopping the batch.

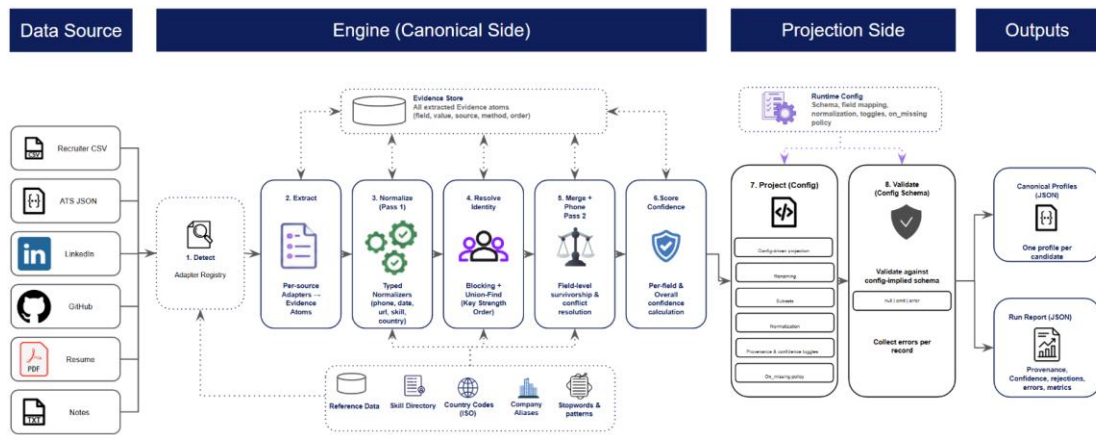


Fig. 1 : Proposed Architecture of North Star (Multi-Source Candidate Data Transformer).

2. CANONICAL SCHEMA & NORMALIZATION

- CanonicalProfile:** candidate_id, full_name, emails[], phones[], location, links, headline, years_experience, skills[], experience[], education[], per field provenance[], field_confidence and overall_confidence.
- Phones:** use E.164 in two steps and resolve +CC or -default-region numbers first. The rest of the numbers will be tried after the merge with the cluster's country. Numbers that could not be resolved will be not reported. No guessing should be performed.
- Dates:** uses year, month, null, rendered as YYYY or YYYY-MM ("2019" never becomes "2019-01"). "Present" → end: null + is_current; ranges close at pinned --as-of; ambiguous months become null and are reported.
- Strings & country:** NFC-normalized at intake and casefolded for comparison. Countries are normalized to ISO-3166 alpha-2 codes; filesystem metadata such as file mtime and listing order never influence output.
- Skills:** casefold → strip versions/punctuation → alias dictionary; misses kept verbatim with canonical: false and reduced confidence, never dropped or force-mapped.
- candidate_id:** sha256 (16 hex) of the cluster's smallest strong identifier, content-derived, stable across reruns.

3. MERGE & CONFLICT RESOLUTION

- Identity matching:** hash blocking + union-find (~O(n)), deterministic and input-order independent. Match-key priority: normalized email (plus-tags and Gmail dots removed) → E.164 phone → canonical GitHub/LinkedIn URL → name + company soft match.
- Safety guards:** a contradiction guard blocks transitive false merges. A multi-identity guard removes keys from files containing multiple identities. Bias: prefer a false split over a harmful false merge.
- Survivorship:** winner by source trust (ATS .90 > CSV .85 > LinkedIn .80 > GitHub .75 > Resume .70 > Notes .50) → extraction reliability (direct 1.0 > regex .90 > dictionary .85 > derived .60) → in-data recency → stable tie-breaker.
- Field resolution:** location resolved atomically from one source; sets unioned after normalization and ranked by confidence; experience merged by company + overlapping dates; conflicting derived values (e.g. years_experience) should be kept as alternatives with reduced confidence.
- Confidence:** evidence strength, $s = trust \times reliability$; $agreement = 1 - \prod(1 - s_i)$ (noisy-OR); $field\ confidence = agreement \times support (\sum agree / \sum all)$. Ordinal, explainable and hand-recomputable, not a calibrated probability; correlated sources may inflate it.

4. RUNTIME CUSTOM-OUTPUT CONFIG

- Config:** output fields as [path, from, type, normalize, required] with provenance/confidence options and on_missing: null | omit | error; path grammar limited to four checked forms (name, a.b, a[b], a[[b]]). Invalid paths, type mismatches or unknown normalizers fail before processing.
- Per-field pipeline:** resolve → normalize (idempotent) → handle missing → type-check; failed normalization counts as missing and is reported. Required + missing ⇒ record error. Bad records are excluded and reported, the batch continues. The output JSON Schema is generated dynamically from the config.

5. DELIBERATELY LEFT OUT & SCALE

- Left out:** live GitHub/LinkedIn fetching (breaks offline determinism; recorded-response adapters fill the slot), LLM extraction (plug-in slot with its own trust weight), phonetic/embedding name matching (false-merge risk outweighs dedupe gain), cross-cultural name parsing (full_name kept whole), config constructs beyond the four (grown on demand).
- Scale:** in-memory, tens of MB at 10^4 candidates; ~O(n) blocking; determinism verified by a shuffled-input byte-compare test.